

Software Engineering

President University Lost Center



Lecturer

Hendra Jayanto, M.Sc

Name:

1. Andri	001202400094
2. Farrel Vidyano sanggen	001202400128
3. Hengky Setiawan	001202400198
4. Michelle	001202400017
5. Naila Olivia Ramadhani	001202400112
6. Reza Fahlevi	001202400066

Study Program: IT Cyber Security Class 3

PRESIDENT UNIVERSITY

Jababeka Education Park, Jl. Ki Hajar Dewantara, Kota Jababeka, Cikarang Baru,
Bekasi 17550 Indonesia

2026

Software Engineering.....	1
I. Requirement Analysis.....	3
I.1 Business Background.....	3
I.2 Problem Statement.....	3
I.3 Project Objectives.....	3
II. Existing System and Needs.....	4
II.1 Existing System.....	4
II.2 Identified Needs.....	4
II.3 Functional Requirements.....	4
II.4 Organization Structure.....	6
II.5 Proposed System Overview.....	6
II.6 System Modeling.....	7
II.6.1 Use case Diagram.....	7
II.6.2 Activity Diagram.....	8
II.6.3 Sequence Diagram.....	9
II.6.4 Class Diagram.....	10
II.7 System Architecture.....	11
II.8 System Flow.....	11
III. Implementation.....	11
III.1 System Implementation.....	11
III.2 User Interface (UI).....	11
III.3 Coding Implementation.....	11
IV. Testing and Evaluation.....	12
IV.1 Testing Method.....	12
IV.2 Test Cases.....	12
IV.3 Testing Results.....	13
IV.4 Evaluation.....	13
V. Conclusion.....	13

I. Requirement Analysis

I.1 Business Background

President University is a large academic institution with multiple faculties, offices, and student activity units. Every day, students, lecturers, and staff move between buildings, increasing the chance of lost personal belongings such as ID cards, wallets, gadgets, books, and accessories.

I.2 Problem Statement

Currently, Lost and Found handling is done manually and creates several critical issues:

- 1) Manual & Informal Reporting: Items are reported through scattered channels like WhatsApp groups, Instagram stories, or local security offices, making it hard to track.
- 2) Data Storage: Because reports are stored locally in each building (local storage), there is no centralized data.
- 3) Time-Consuming Compilation: Compiling a report of all lost items is difficult and slow because the data is not digital.
- 4) No Verification Platform: There is no official system to verify if the person claiming the item is the real owner, leading to potential fraud.

I.3 Project Objectives

To develop a centralized Lost & Found web-based system that:

- 1) Allows users to report lost and found items.
- 2) Helps users search items efficiently.
- 3) Enables administrators to manage reports.

II. Existing System and Needs

II.1 Existing System

Currently, every building (Branch) works alone.

- 1) Each faculty or branch handles lost items independently
- 2) Reports are stored locally or not recorded at all
- 3) No integration between departments

II.2 Identified Needs

- 1) Central Database: Accessible by all branches in real-time.
- 2) Online Reporting System: Students can input data from anywhere.
- 3) Search Function: To filter items by category or location.
- 4) Role-Based Access: Different logins for Student (User), Security (Admin), and Head Office (Super Admin).

II.3 Functional Requirements

The functional requirements explain the key functionalities that the Campus Connect Lost & Found system should offer. These requirements detail the interaction between the users and the system and the reaction of the system to the interactions.

- 1) User Registration
The system will enable new users (students and staff) to create an account by entering the required personal information like name, email, and password.
- 2) User Authentication and User Login
The system enables registered users to log-in safely with their credentials. Regulated access to limited features like reporting and claiming items is authenticated.
- 3) Browse Item (Guest Access)
The system shall allow unregistered users (guests) to browse and view the list of lost and found items without logging in.
- 4) Report Found Item
The system enables registered users or administrators to report found items where all the information such as description, location, image is provided.

5) Search and Filter Items

The system also offers a search tool through which the user can search items by using key words. The system also allows filtering by category, location and date.

6) View Item Details

The system enables one to see detailed information on all items such as description, pictures, date and location.

7) Claim Item

The system enables registered users to make a claim request of any found item. To confirm ownership of the item, users will be required to supply verification information.

8) Chat / Messaging Feature

The system allows users to communicate with each other through chat features after a claim is initiated. This feature supports direct interaction for verification purposes.

9) Notification System (Web Notification)

The system shall generate real-time web notifications when important events occur, such as new messages, claim submissions, and status updates.

- a) New message
- b) Claim submissions
- c) Status updates

10) Email Notification Integration

The system will automatically notify users by emailing them via an external email service in case of any essential activities and in this way the user will be informed even when they are not online.

11) Centralized Data Storage

All data shall be stored in a centralized database that is accessible to all branches in real time.

II.4 Organization Structure

To make this project, our team is divided into these roles

- 1) Project Manager (PM): The leader. Check the deadline and make sure everyone works.
- 2) System Analyst: The planner. Studies the problem and decides what features we need.
- 3) UI/UX Designer: Designs website interface and user experience (buttons, colors, layout)
- 4) Backend Developer: Handles database and server-side logic
- 5) Frontend Developer: Implements user interface
- 6) Testing & Quality Assurance: The checker. Tests the website to find bugs/errors before we release it.

II.5 Proposed System Overview

- 1) System overview

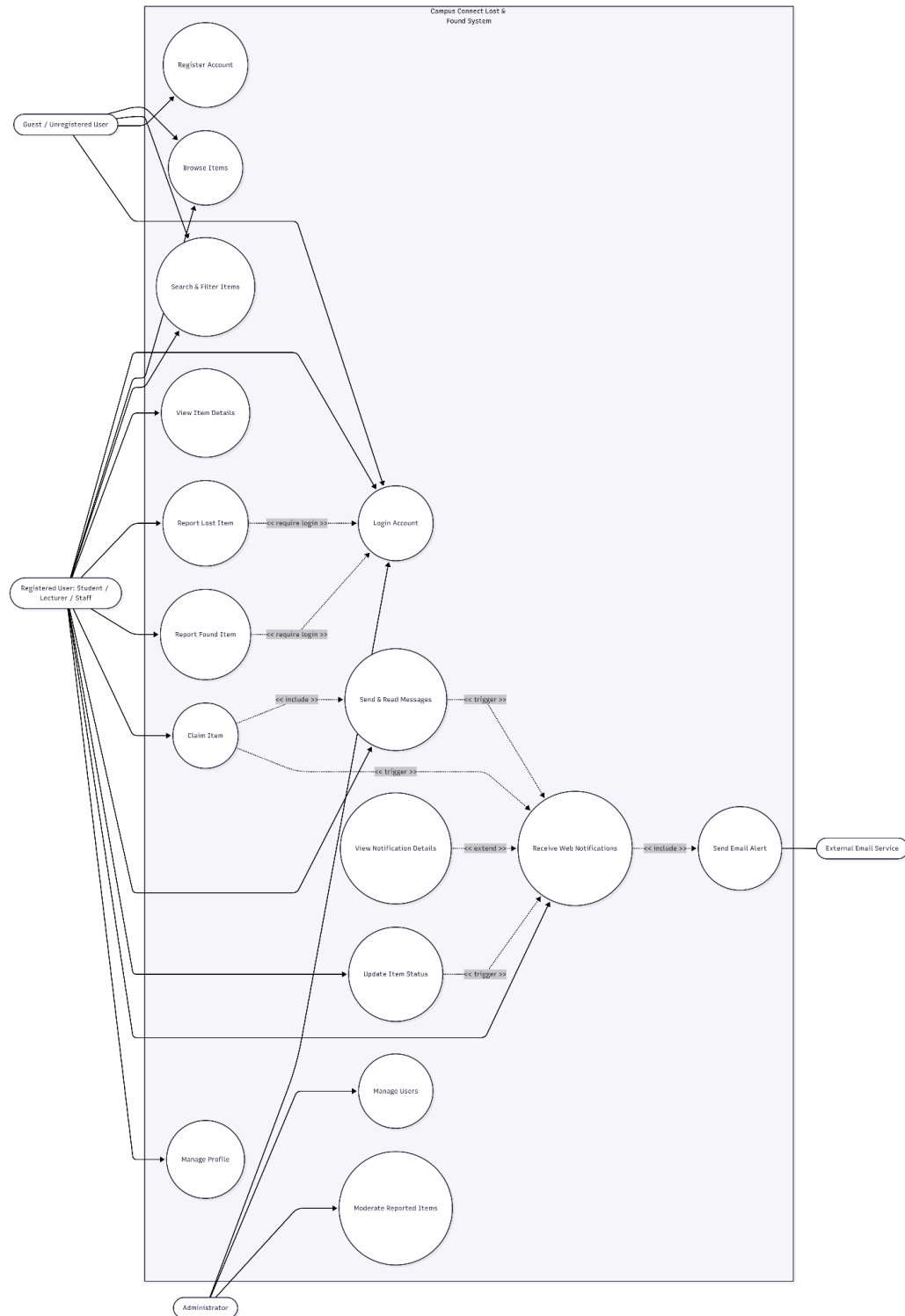
The suggested solution is a web-based Lost and Found system that aims to streamline all the operations of reporting and tracking lost items in President University. The system enables users to give reports on lost and found, search items and make claims.

- 2) How system works

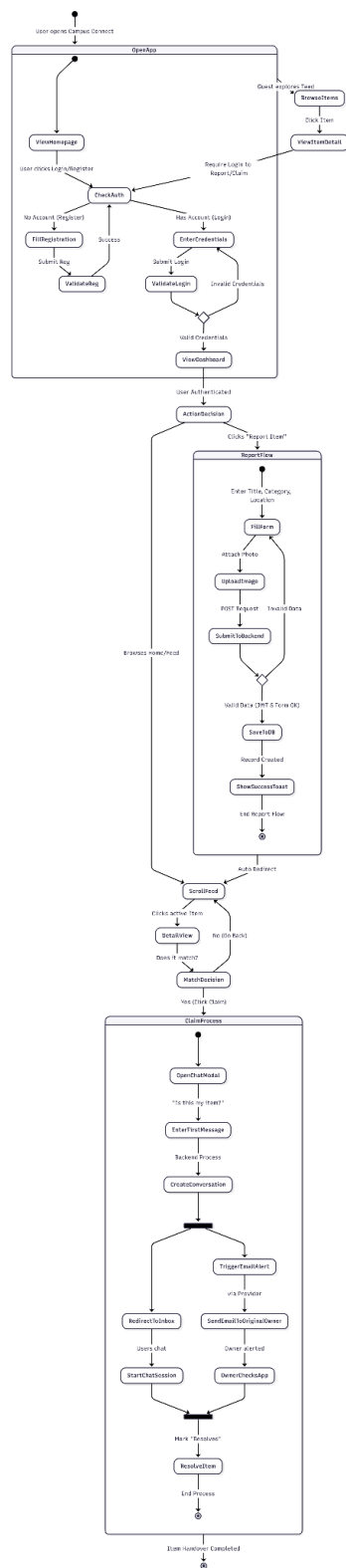
The system operates by keeping all reports in the central database which can be accessed by all the users in the different branches. By applying a claim, the system allows a chat feature as a user can communicate directly with each other to verify a claim. The system also offers real-time notification and email alerts so that users can be aware of any action concerning their reports or claims.

II.6 System Modeling

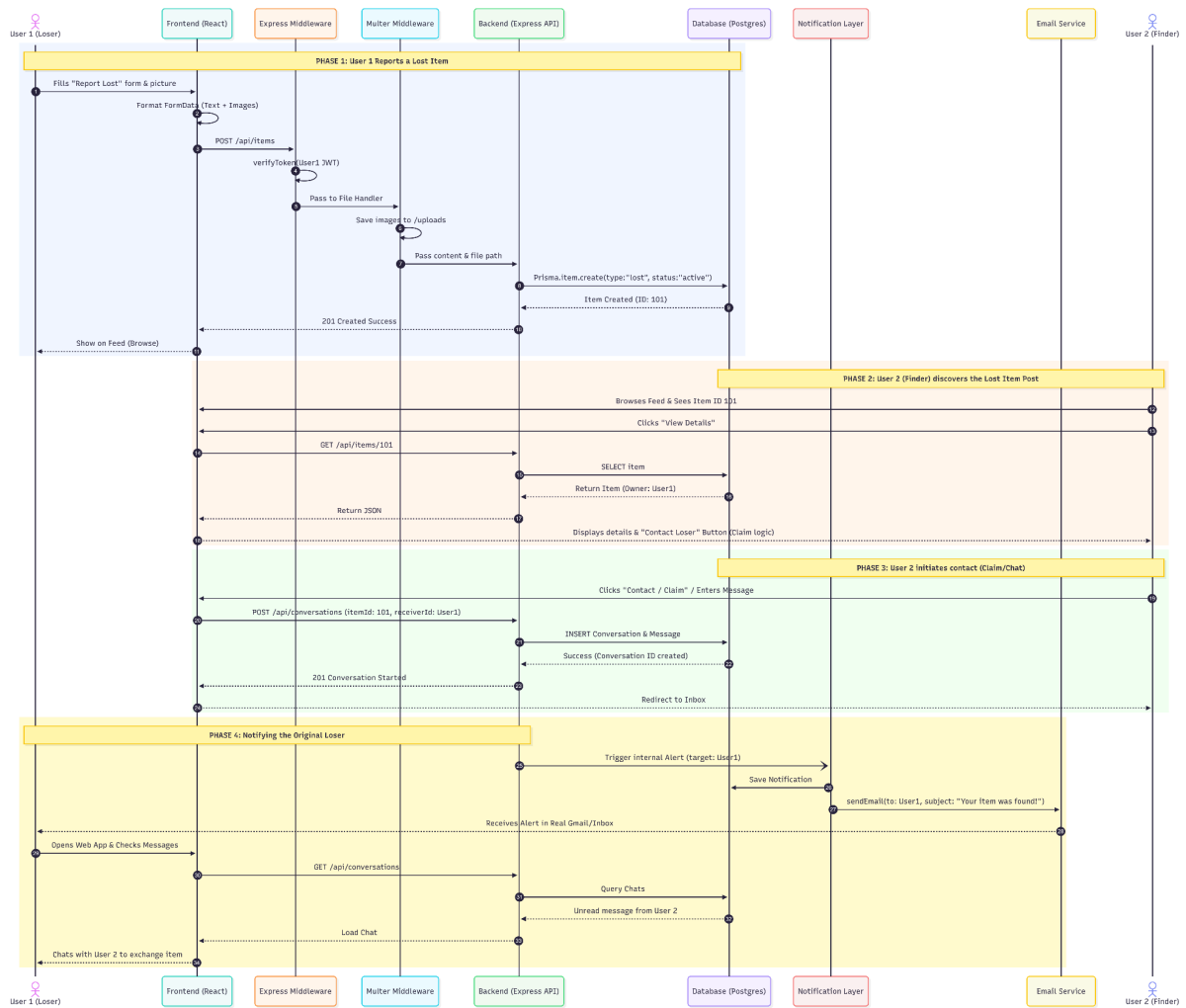
II.6.1 Use case Diagram



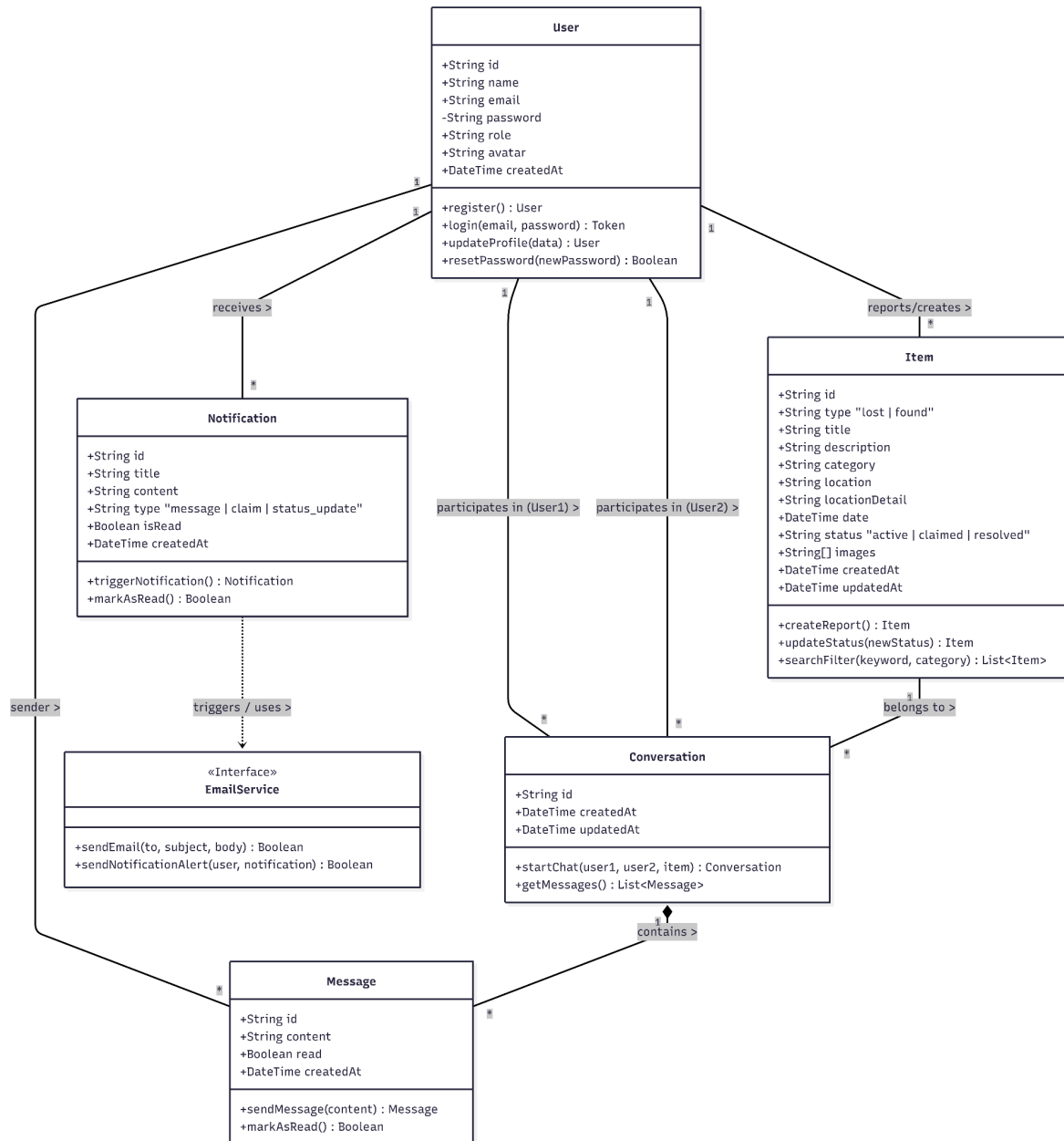
II.6.2 Activity Diagram



II.6.3 Sequence Diagram



II.6.4 Class Diagram



II.7 System Architecture

- 1) Frontend: React.js, React Router, Framer Motion,
- 2) Database: postgresql
- 3) Hosting: Vercel
- 4) Development Tools: VS Code, etc.

II.8 System Flow

The system flow of the President University Lost Center system is described as follows:

- 1) The user accesses the website
- 2) The user can browse items without logging in
- 3) To access advanced features, the user must register and log in
- 4) The user can report lost or found items
- 5) The system stores the data in a centralized database
- 6) Other users can search and view item details
- 7) A user can submit a claim for a found item
- 8) The system enables a chat feature between users
- 9) Admin verifies the claim request
- 10) The system sends notifications and email updates

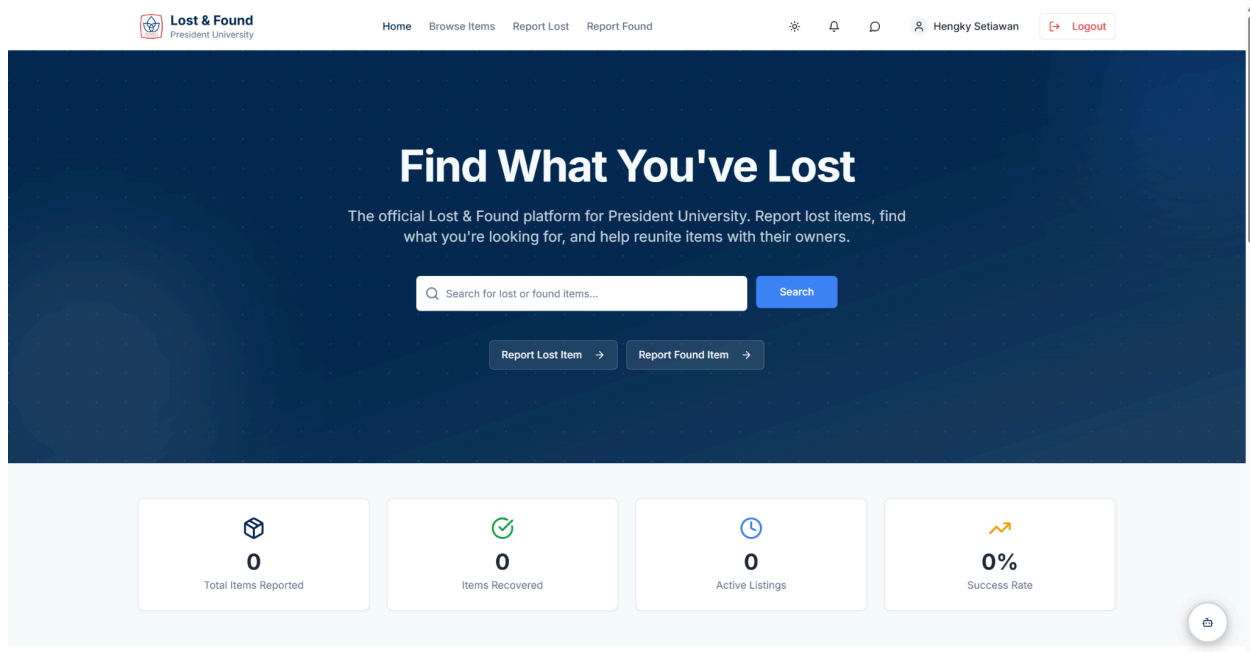
III. Implementation

III.1 System Implementation

The system implementation phase involves developing the Lost & Found web-based application based on the system design defined in the previous chapter. The system is designed to help students and staff of President University report lost items, find items, and reconnect items with their owners efficiently.

The application is implemented as a web based system using a client-server architecture, where users interact with the system through a web browser, while the server processes requests, handles business logic, and manages data storage.

III.2 User Interface (UI)



Home page interface

The Home Page serves as the main entry point of the Lost & Found system. It is designed to provide users with quick access to the system's core functionalities, including searching for items and reporting lost or found items.

At the top of the page, there is a **navigation bar** that contains the system logo and menu options such as *Home*, *Browse Items*, *Report Lost*, and *Report Found*. On the right side, user-related features are displayed, including user profile information (username), notification icon, settings, and a logout button. This layout allows users to easily navigate through the system.

The main section of the homepage features a **hero banner** with a prominent heading, "*Find What You've Lost*", which clearly communicates the purpose of the system. Below the heading, there is a short description explaining the platform's function in helping users report and find lost items.

A **search bar** is placed at the center of the page, allowing users to quickly search for lost or found items using keywords. Next to the search field is a **Search button**, which triggers the search functionality.

Below the search bar, there are two primary action buttons:

- **Report Lost Item**
- **Report Found Item**

These buttons are designed to provide quick access to the reporting features, improving user efficiency.

Further down, the homepage displays a set of **statistical summary cards**, which include:

- Total Items Reported
- Items Recovered
- Active Listings
- Success Rate

Each card presents key system data in a simple and visually clear format, helping users understand the current status of reported items.

The interface uses a **clean and modern design** with a blue color theme, consistent typography, and well-organized layout. The design also ensures responsiveness, allowing the system to be accessed from different devices such as desktops, tablets, and smartphones.

Users can access the system through the following link:

[PULC Website](#)

IV. Testing and Evaluation

IV.1 Testing Method

The testing method used in this project is Black Box Testing. This method focuses on testing the system based on its features and functions such as registration, login, reporting items, searching, claiming items, messaging, and notifications. The goal is to make sure that every feature works correctly according to the system requirements.

IV.2 Test Cases

No	Feature	Test Case	Expected Result	Actual Result	Status
1	Registration	User inputs valid data	Account is created	Account created successfully	Pass ✓
2	Login	Correct email and password	User can log in	Login successful	Pass ✓
3	Login	Input wrong email or password	Error message appears	Error message shown	Pass ✓
4	Browse Item	Users opens item list	Items are displayed	Items displayed	Pass ✓
5	Report Item	User submit item report	Item is saved in system	Item displayed	Pass ✓
6	Search	User searches "Powerbank"	Related item appear	Item displayed correctly	Pass ✓
7	View Item Detail	User click item	Item details are shown	Item displayed	Pass ✓
8	Claim Item	User submit claim	Claim is sent	Claim submitted	Pass ✓
9	Messaging	User sends message	Message is delivered	Message sent	Pass ✓
10	Notification	New claim created	Notification	Notification	Pass ✓

			appears	shown	
11	Profile	User views profile	User data displayed	Profile works correctly	Pass ✓

IV.3 Testing Results

Based on the testing, all main features of the system work properly. Without significant failures, the system is capable of user registration, login, item reporting, searching, claiming and messaging. Alerts are also functional and assist users to be notified. There were also some minor problems, including slow loading of some pages and plain validation messages.

IV.4 Evaluation

The system has successfully solved the main problems of the old manual system. Before this system, lost and found items were reported in different places, making it hard to track. Now, the system provides a centralized platform, easier item searching, better communication between users. The verification and messaging features also improve security and trust between users. In conclusion, the system is effective and helpful for managing lost and found items in President University.

V. Conclusion

In conclusion, the President University Lost Center system can effectively manage the issues that have been noted in the current manual system. The system enhances lost-item searching, reporting and management efficiency by means of a centralized web-based platform, This system is expected to improve efficiency, transparency, and security in handling lost and found items within the university environment..

Security is also increased by the verification features and the user experience is guaranteed by real-time notifications and communication tools. In general, the suggested system offers an effective and stable method of controlling lost and found items in President University.